

SUPPLEMENTARY MATERIAL

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1. ADDITIONAL ACOUSTIC IMAGING DETAILS

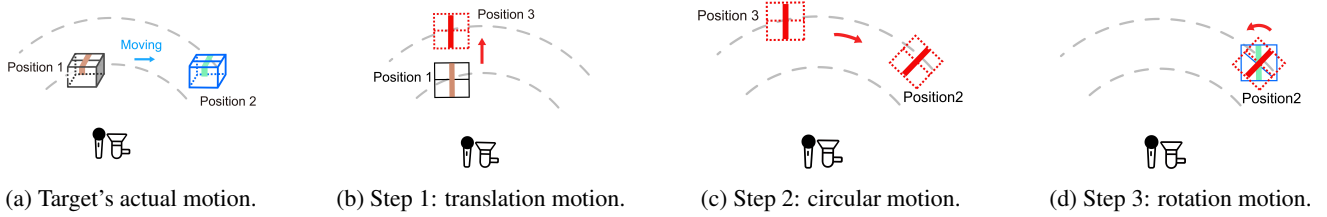


Fig. 1: The target's actual motion can be decomposed into three steps: (a) translation motion, (b) circular motion, and (c) rotation motion.

1.1. The principle of ISAR

Inverse Synthetic Aperture Radar (ISAR) is commonly employed in military applications to identify the shapes of moving targets, such as ships. Unlike motion tracking systems that consider the target as a single point, ISAR regards the target as an array of scatter points. ISAR imaging converts the target's translational motion into rotational motion, allowing for the capture of reflected signals from various perspectives around the target.

As shown in Fig.1a, consider a target moving from Position 1 to Position 2. This movement is illustrated in three stages in Fig.1b-1d. First, the target moves radially from Position 1 to Position 3 (Fig.1b), resulting in additional propagation delays in the signals reflected from each scatter point. These delays introduce image distortion, which is corrected using our echo alignment technique. In the second stage, the target moves along a circular path centered on the imaging device, from Position 3 to Position 2 (Fig.1c), without affecting propagation delays or the final image outcome. In the final stage, the target rotates around its geometric center by a specific angle (Fig. 1d), aligning with the azimuth angle traced during its actual movement.

1.2. The Basics of Chirp Signal

A speaker emits a series of chirp signals with frequencies that linearly sweep over time, as shown in Fig. 2a. The frequency of each chirp ranges from $f_c - \frac{B}{2}$ to $f_c + \frac{B}{2}$, where f_c is the carrier frequency and B is the bandwidth. Each chirp lasts for a duration T_c , followed by a pause, T_e , to allow all echoes from one chirp to return before the next chirp is emitted. Each chirp can be mathematically expressed as

$$s(t) = \text{rect} \left[\frac{t}{T_c} \right] \cos \left(2\pi \left(f_c t + \frac{k}{2} t^2 \right) \right), \quad (1)$$

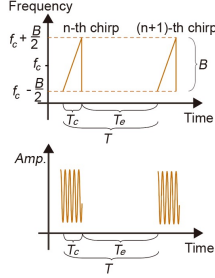
where $k = \frac{B}{T_c}$ is the slope of the chirp. The term $\text{rect}[\cdot]$ refers to the rectangle window function, which is defined as

$$\text{rect}[\cdot] = \begin{cases} 1 & |\cdot| \leq 0.5 \\ 0 & |\cdot| \geq 0.5 \end{cases} \quad (2)$$

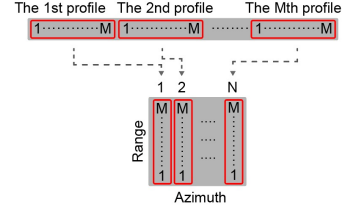
The total duration of each chirp cycle, including the pause, is $T = T_c + T_e$. The transmitted chirp signal is reflected by the target and received by a microphone, creating an echo. The time delay ($\Delta\tau$) of the echo is typically calculated by correlating the transmitted chirp signal with the received echo [1]. Using this time delay, the distance D between the target and the device can be determined by:

$$D = \frac{\Delta\tau \times v_s}{2}, \quad (3)$$

where v_s is the sound propagation speed in the air.



(a) The chirp signal.



(b) The profile matrix.

Fig. 2: The signal illustration for imaging.

1.3. Signal Model for Imaging

The received echo signals are organized into a profile matrix for imaging, as illustrated in Fig. 2b. This matrix arranges M samples per column with N consecutive collected profiles to construct the image. Then we apply the Hilbert transform to the echo signals, followed by dechirping each column to extract the intermediate frequency (IF) signal matrix. Subsequently, a 2-D Fast Fourier Transform (FFT) is performed on the IF signal matrix along both the range and azimuth dimensions. This step effectively maps each scatter point in physical space to a corresponding pixel in the image space, I , where a scatter point p is identified by a peak at position (p_r, p_a) in the 2-D image space, which can be represented as

$$I(i_a, i_r) = A \text{sinc} \left[\frac{2kT}{v_s} (i_r - p_r) \right] \times \text{sinc} \left[\frac{2\theta}{\lambda} (i_a - p_a) \right], \quad (4)$$

where A is a constant related to the amplitude and chirp duration, v_s is the sound speed, λ is the wavelength, θ is the angle that the target rotates during the imaging process, i_r and i_a are the indexes along the range dimension and azimuth dimension, respectively.

2. ADDITIONAL REGISTRATION DETAILS

2.1. Definition of the 16 Human Joints

Fig. 3 shows the 16 human joints we considered.

2.2. Why Soft-Argmax

Soft-argmax provides a smooth and differentiable alternative to hard argmax for mapping a 2D joint feature to the canonical 3D space. For each modality $m \in \{a, v\}$, we compute the cosine similarity between the joint feature z_j^m and the embedding $h_i = g(c_i)$ of each canonical candidate location $c_i \in \mathcal{C}$. The similarities are normalized by softmax to obtain correspondence weights w_{ji}^m over the N candidate locations.

Rather than selecting only the candidate with the largest similarity, soft-argmax computes the registered 3-D location as the softmax-weighted expectation

$$\hat{K}_j^m = \sum_{i=1}^N w_{ji}^m c_i. \quad (5)$$

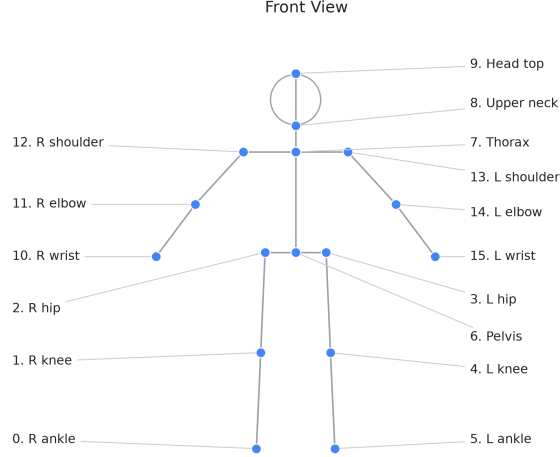


Fig. 3: The 16 critical joints used in our registration module. The front view shows the primary joints, including the head, shoulders, elbows, wrists, hips, knees, and ankles.

This formulation is fully differentiable, allowing the registration loss to propagate gradients through the correspondence weights and the modality-specific feature extractors. It also avoids the discontinuity of hard nearest-candidate selection.

2.3. Acoustic Joint Localization

First, we employ Fast-CNN [2] to detect the region of humans in acoustic images. Fast-CNN uses a bounding-box technique to efficiently scan acoustic images, identifying regions where shape and intensity patterns suggest human presence. This initial detection stage provides coarse spatial localization of human figures within the acoustic image.

Second, to achieve robust pose estimation in human regions, we employ a Gaussian Mixture Model (GMM). GMM is particularly well-suited for this task because it models the underlying uncertainty and noise commonly found in acoustic data through a stochastic framework. Unlike deterministic clustering methods, GMM provides more stochasticity, allowing it to handle ambiguous boundaries between body parts gracefully. Furthermore, it naturally captures the spatial clustering of anatomical structures and implicitly encodes the relative positions and orientations of different segments, making it an effective tool for refining human body pose estimation.

We then locate 16 key joint points, each constrained to its corresponding segmented region as detected by the GMM. To locate each joint, we calculate the geometric center of each segment, $\bar{x} \triangleq \sum_{i=1}^N x_i / N$, where x_i are points within a segment and N is the total number of points in the segment, and analyze areas of maximal density as potential joint locations. This selection focuses on intersections and peripheries of these segments, minimizing the Euclidean distance $d(\mathbf{p}, \mathbf{q}) \triangleq \sqrt{\sum_{i=1}^N (q_i - p_i)^2}$ between centroids $\mathbf{p}$ and $\mathbf{q}$ of adjacent segments, as well as density measurements to identify clusters of data points within each segment.

See Fig. 4 for an illustrative example on acoustic joint localization.

3. MODALITY MASKING AND GLOBAL-LOCAL FUSION

Reliable 3D human mesh reconstruction requires consistent performance across diverse environments. While RGB and acoustic signals offer complementary strengths - RGB providing detailed features in good lighting and acoustic signals excelling in low visibility - naively combining these modalities creates a critical challenge. During training, models naturally gravitate toward the higher-quality RGB features in well-lit conditions [3, 4]. This bias manifests in two ways: the model fails to fully leverage acoustic signals when they would be most valuable (such as in poor lighting or occlusions), and simultaneously becomes overly reliant on RGB data quality, making the system vulnerable to environmental variations.

Understanding this fundamental challenge motivates our key technical contribution: strategies to enforce balanced learning from both modalities. By preventing over-reliance on either modality during training, we can ensure the system maintains robust performance regardless of environmental conditions, effectively utilizing whichever signal provides the most reliable information in any given situation.

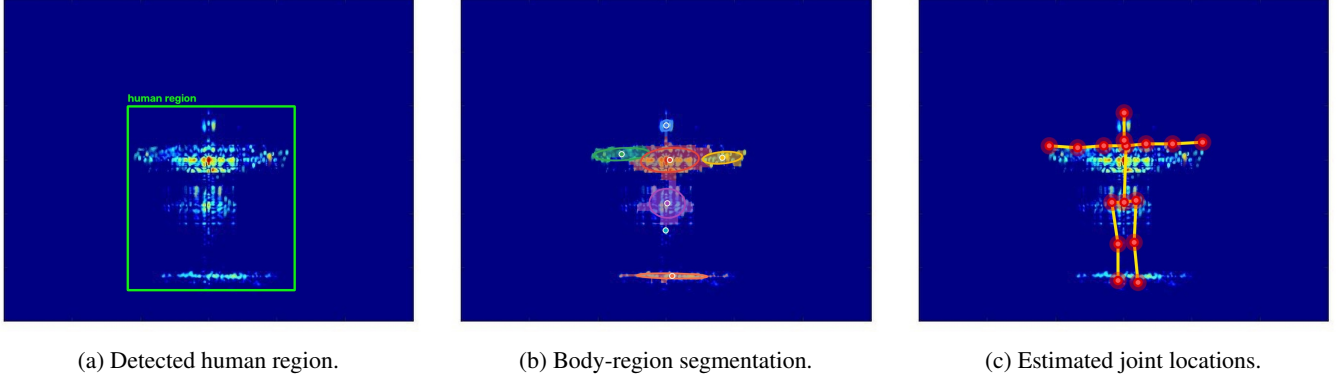


Fig. 4: Illustration of acoustic joint localization: (a) human-region detection, (b) body-region segmentation, and (c) joint localization.

To ensure balanced learning across modalities, we implement a two-pronged approach. First, we employ modality masking, which randomly occludes either RGB or acoustic inputs during training. This forces the model to learn robust features from each modality independently, preventing over-reliance on any single input source. Second, we incorporate a Global Integrated Module (GIM), inspired by [4], which dynamically weights the contribution of each modality based on input quality. While modality masking ensures independent feature learning, GIM optimizes the fusion of these features by adaptively combining information according to the reliability of each source. Full details of the GIM architecture and training process can be found in [4].

4. IMPLEMENTATION DETAILS

4.1. Training Setup

All models are implemented in Pytorch and trained on an NVIDIA GeForce RTX 4090. Each network is trained from scratch using the Adam optimizer with an initial learning rate of 0.2. The learning rate decays to 90% of its previous value every 10 epochs, with a lower bound of 0.005.

4.2. Acoustic device and configuration

We implement our proof-of-concept system using a commodity device: a Bela [5] platform. Signal processing and analysis are carried out using MATLAB.

Bela Platform: The Bela [5] platform is favored in acoustic sensing research for its versatility in supporting various microphone and speaker configurations. We equipped the Bela with a PUI Audio AS03104MR-N50-R speaker for signal transmission and a SparkFun BOB-18011 MEMS microphone for signal reception.

Acoustic Signals: To avoid interference from environmental noise, which typically falls below 16 kHz [6], we configure our system to sweep frequencies from 18 kHz to 22 kHz . The duration of each chirp is set as 1 ms , and the empty duration is set as 5 ms .

4.3. Dataset Details

For our dataset of acoustic images, we defined eight common daily actions: standing, arms spreading, raising one hand, raising both hands, arms spread diagonally upward, arms stretched downward, arms extended straight up, and raising one arm to the side. For each action, data was collected from 20 volunteers (10 male and 10 female) with 100 samples per action. We introduced 3 challenges—poor lighting, occlusions, and smoke (to simulate NLOS)—each representing 25% of the total collected data. Fig 5 shows these poses and their corresponding generated acoustic images. We used a calibrated Samsung S24 as the camera system to capture RGB images, which serve as the ground truth in our dataset.

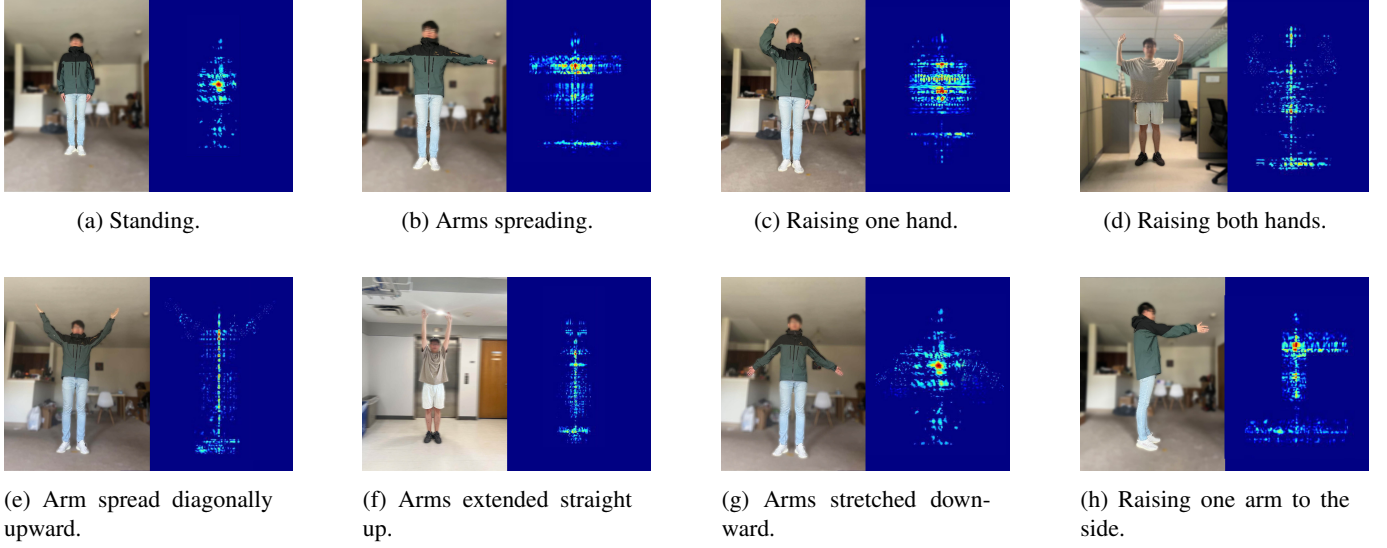


Fig. 5: The signal illustration for imaging.

5. REFERENCES

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